Renewable Energy Sources

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Introduction to Renewable Energy

Renewable energy is energy obtained from natural sources that are replenished constantly, such as sunlight, wind, flowing water and heat from the Earth. Unlike fossil fuels like coal, oil and natural gas, which took millions of years to form and are being used up, renewable sources will not run out on human time scales and produce little or no greenhouse gas emissions or air pollution when generating power. Renewable energy is therefore central to fighting climate change and improving energy security. In 2023 and 2024, renewables supplied around 30 percent of the world's electricity, and solar power was the fastest-growing source.

Five Renewable Energy Sources

Five important renewable energy sources are solar, wind, hydropower, biomass and geothermal energy. Tidal and ocean energy are also renewable.

Solar energy converts sunlight directly into electricity using photovoltaic panels, or concentrates sunlight to produce heat in solar thermal plants, and also heats water in rooftop solar water heaters.

Wind energy uses the kinetic energy of moving air to turn the blades of a wind turbine connected to a generator. Wind farms can be onshore or offshore. A modern turbine can have blades more than 80 metres long.

Hydropower uses the energy of flowing or falling water, usually stored behind a dam, to spin turbines. It is the largest renewable source of electricity globally. Examples in India include Bhakra Nangal, Tehri and Sardar Sarovar, and Tamil Nadu's Mettur and Kundah projects. Small hydro and pumped-storage plants also provide storage.

Biomass and bioenergy uses organic matter such as agricultural waste, wood, cow dung and food waste to produce heat, electricity or biofuels such as ethanol and biogas. India's ethanol blending programme aims at blending ethanol with petrol, reaching 20 percent (E20) in 2025.

Geothermal energy uses heat from beneath the Earth's surface, available as steam or hot water, to generate power and heat. Iceland and the Philippines make major use of it, and India has potential at Puga Valley in Ladakh. Tidal energy uses the rise and fall of the sea, as at the La Rance plant in France, and wave energy uses the motion of waves.

Advantages of Renewable Energy

Renewable energy cuts carbon emissions and pollution, reduces dependence on imported fuel, creates jobs, provides electricity to remote villages through off-grid systems, and has costs that have dropped sharply: the cost of solar panels has fallen by about 90 percent in a decade. The fuel, sunlight or wind, is free.

Limitations

Solar and wind are intermittent, meaning they depend on weather and time of day, so grid stability requires energy storage such as batteries and pumped hydro, backup power and flexible grids. Large hydro projects can displace communities and affect rivers. Renewable plants need land and materials, and the recycling of panels, blades and batteries is a growing concern.

Renewable Energy in India

India is the world's third-largest producer and consumer of electricity and is among the leaders in renewable capacity. India has set a target of 500 gigawatts (GW) of non-fossil electricity capacity by 2030, and in 2025 non-fossil sources crossed 50 percent of installed capacity, years ahead of the 2030 target in India's climate pledge. Solar capacity crossed 100 GW in early 2025. The country has the world's largest solar parks, such as Bhadla in Rajasthan, and the International Solar Alliance, co-founded by India and France in 2015, is based in Gurugram.

Which Indian State Leads in Wind Power?

Tamil Nadu is India's wind power pioneer and has long been the leading state in installed wind power capacity, with roughly 10,000 to 12,000 megawatts, though Gujarat has recently been very close and the ranking can change as new projects are added. The wind farm at Muppandal near Kanyakumari, with a capacity of over 1,500 MW across thousands of turbines, is among the largest onshore wind clusters in India. The strong winds of the Palghat Gap and the southwest monsoon season make the state especially suitable. Other leading wind states are Gujarat, Karnataka, Maharashtra and Rajasthan. India has the fourth-largest installed wind capacity in the world, and offshore wind projects are planned off Tamil Nadu and Gujarat.

What Is Green Hydrogen?

Green hydrogen is hydrogen gas produced by splitting water into hydrogen and oxygen through electrolysis, using electricity from renewable sources such as solar or wind. Because no fossil fuels are used, producing green hydrogen emits no carbon dioxide. By contrast, grey hydrogen is made from natural gas, releasing carbon dioxide, and blue hydrogen is made from natural gas with the carbon dioxide captured and stored. Hydrogen is a clean fuel, because burning it or using it in a fuel cell produces only water. It can decarbonise hard-to-electrify sectors such as steel, fertilisers, shipping and heavy transport, and can also store energy. India launched the National Green Hydrogen Mission in January 2023, with an outlay of about Rs. 19,744 crore, aiming at producing 5 million tonnes of green hydrogen a year by 2030. Challenges include high cost, the need for large amounts of renewable power and safe storage and transport.

Challenges and Future

Future trends include cheaper batteries, floating solar farms, offshore wind, smart grids, green hydrogen, and rooftop solar for households under the PM Surya Ghar scheme.

Quick Facts

- Five renewable energy sources are solar, wind, hydropower, biomass and geothermal.
- Tamil Nadu is the pioneer and traditional leader in wind power in India, with Gujarat close behind.
- Green hydrogen is made by electrolysis of water using renewable electricity.
- India's National Green Hydrogen Mission was launched in January 2023.
- India's target is 500 GW of non-fossil electricity capacity by 2030.